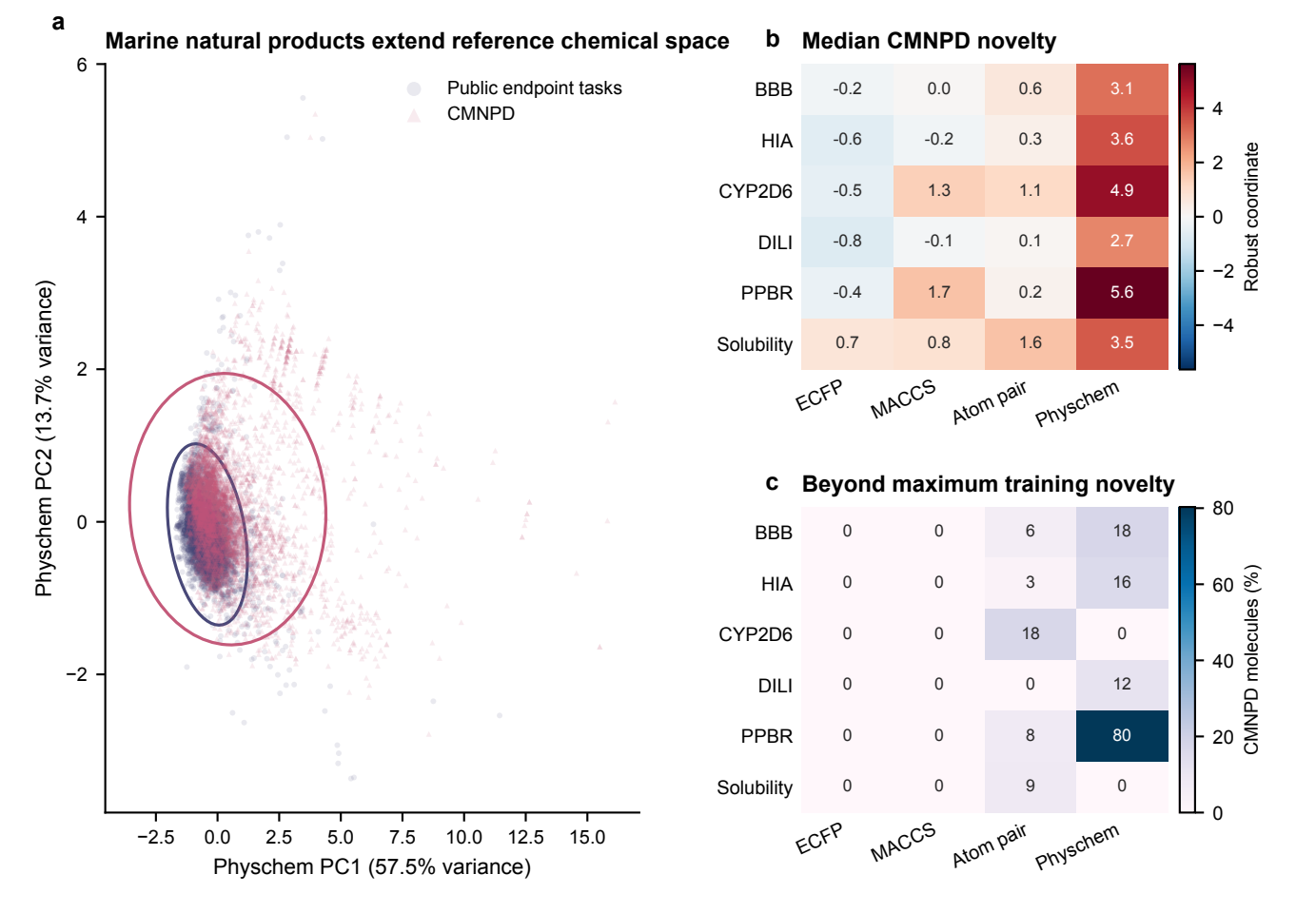

a
Marine natural products extend reference chemical space
Public endpoint tasks
CMNPD
Physchem PC2 (13.7% variance)
Physchem PC1 (57.5% variance)
6
4
2
0
−2
−2.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0
b
Median CMNPD novelty
BBB -0.2 0.0 0.6 3.1
HIA -0.6 -0.2 0.3 3.6
CYP2D6 -0.5 1.3 1.1 4.9
DILI -0.8 -0.1 0.1 2.7
PPBR -0.4 1.7 0.2 5.6
Solubility 0.7 0.8 1.6 3.5
ECFP
MACCS
Atom pair
Physchem
Robust coordinate
4
2
0
−2
−4
c
Beyond maximum training novelty
BBB 0 0 6 18
HIA 0 0 3 16
CYP2D6 0 0 18 0
DILI 0 0 0 12
PPBR 0 0 8 80
Solubility 0 0 9 0
ECFP
MACCS
Atom pair
Physchem
CMNPD molecules (%)
80
60
40
20
0